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(54) **TRANSFORMED MICROORGANISM AND METHOD OF PRODUCING POLYHYDROXYALKANOATE**

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CPC **C12N 9/1029** (2013.01); **C12P 7/625** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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(57) **ABSTRACT**

Provided is a transformed microorganism that has a polyhydroxyalkanoate synthase gene and in which expression of a minD gene is enhanced. Also provided is a transformed microorganism that has a polyhydroxyalkanoate synthase gene and in which expression of a minC gene and a minD gene is enhanced. In this transformed microorganism, expression of a minE gene may be enhanced or reduced. Also provided is a method of producing a PHA, the method including the step of culturing any of the transformed microorganisms in the presence of a carbon source.

20 Claims, 4 Drawing Sheets

Specification includes a Sequence Listing.

Fig. 1

Comp. Example 1

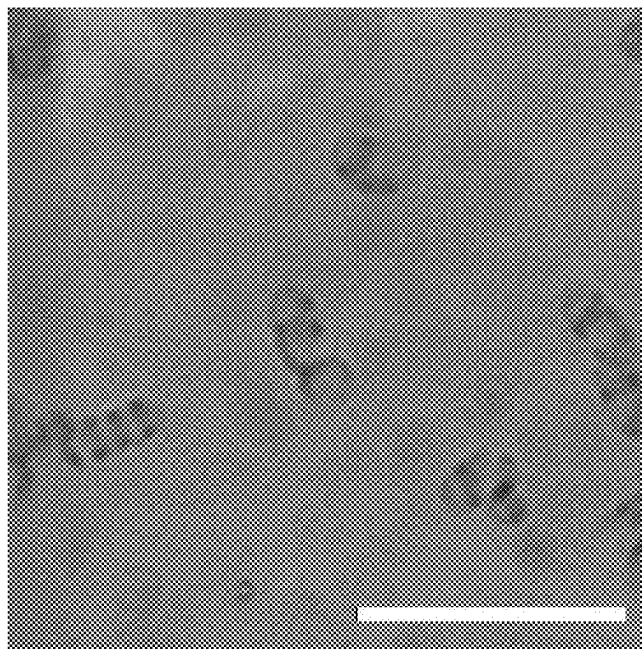


Fig. 2

Comp. Example 2

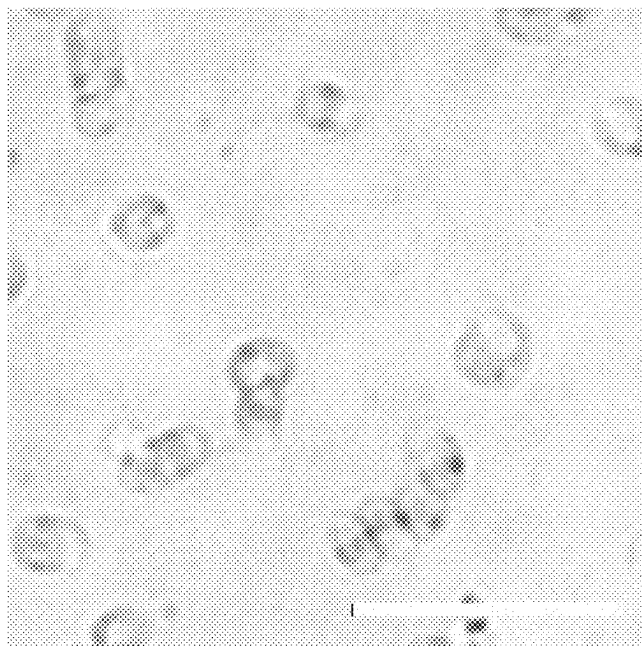


Fig.3

Comp. Example 3

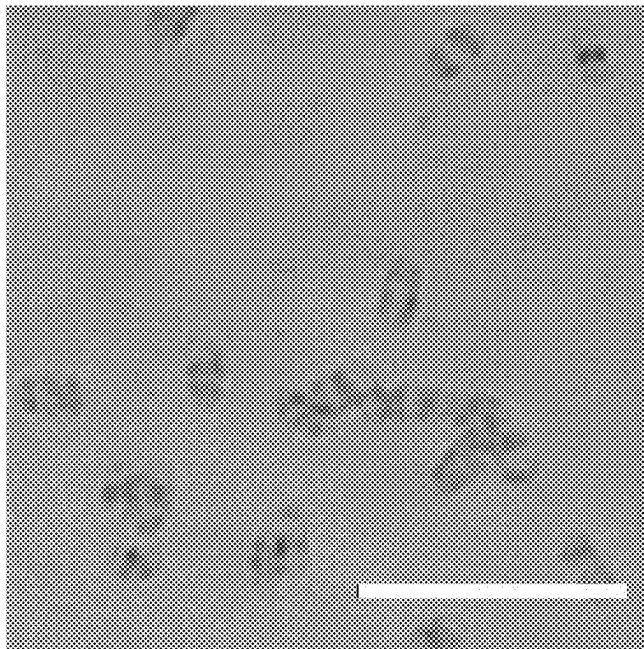


Fig.4

Comp. Example 4

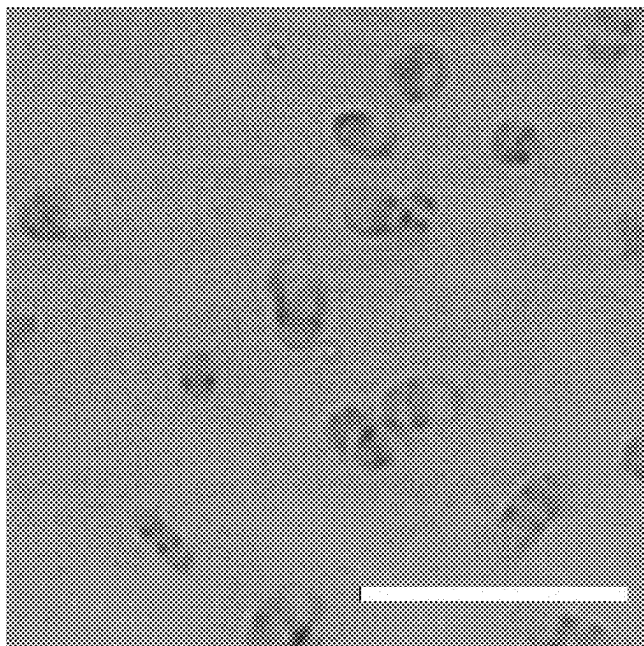


Fig.5

Example 1

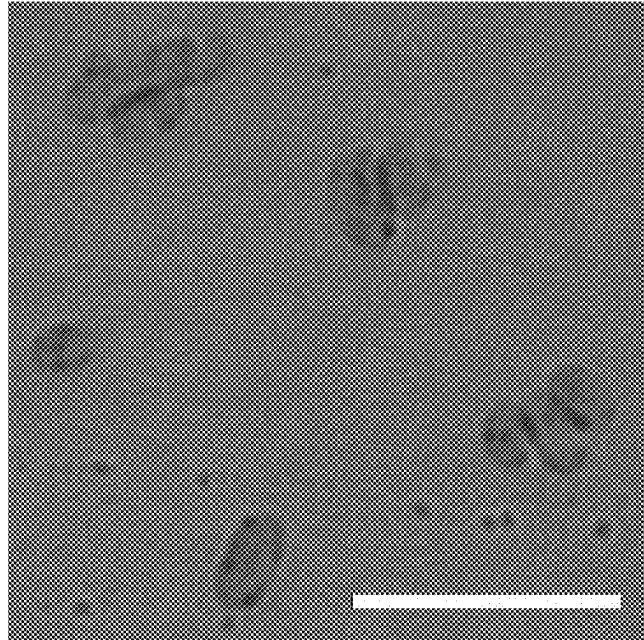


Fig.6

Example 2

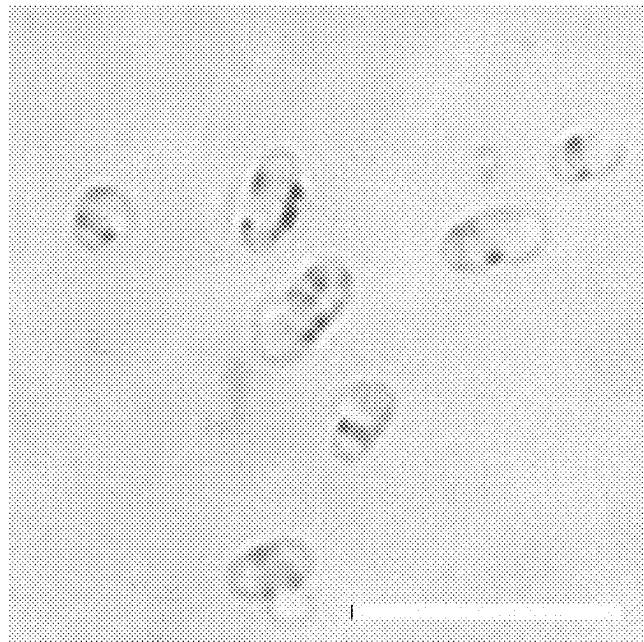


Fig.7

Example 3

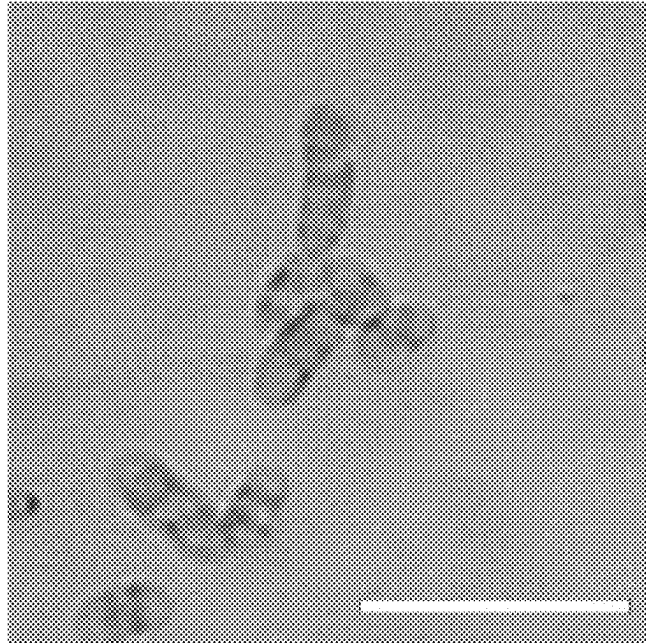
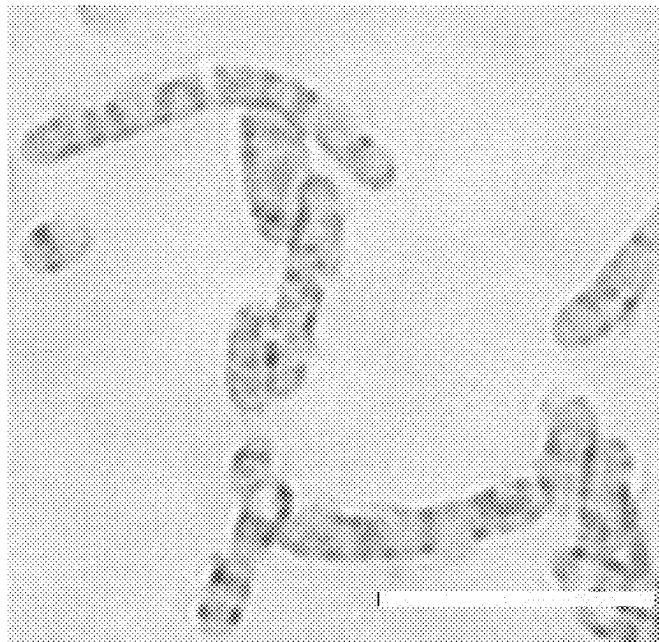


Fig.8

Example 4



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TRANSFORMED MICROORGANISM AND METHOD OF PRODUCING POLYHYDROXYALKANOATE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International Application No. PCT/JP2020/003156, filed Jan. 29, 2020, and claims priority to Japanese Application No. 2019-036006, filed Feb. 28, 2019, and Japanese Application No. 2019-036008, filed Feb. 28, 2019, the disclosures of all of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

The present invention relates to a transformed microorganism capable of elaborating a polyhydroxyalkanoate and a method of producing the polyhydroxyalkanoate using the transformed microorganism.

BACKGROUND ART

There is a growing awareness of environmental issues, food issues, health, and safety, and more and more people are becoming nature-oriented. Against such a background, material production using microorganisms (such as fermentative production and bioconversion) is becoming increasingly significant and important. Microbial material production is applied also to production of protein pharmaceuticals and production of nucleic acids for gene therapy. For example, ethanol production, acetic acid production, and medical protein production using microorganisms such as yeasts and bacteria are actively employed industrially.

An example of the microbial material production is microbial production of polyhydroxyalkanoates (occasionally referred to as "PHAs" hereinafter) which are considered promising biodegradable plastics for industrial use (see Non Patent Literature 1). PHAs are thermoplastic polyesters produced and accumulated as energy storage materials in cells of many kinds of microorganisms and are biodegradable. Nowadays, the heightened environmental awareness has led to increasing attention to non-petroleum-based plastics. In particular, there is a strong demand for practical use of PHAs produced and accumulated in microorganisms because such PHAs are absorbed into the carbon circulation process in the nature and are therefore expected to have little adverse impact on the ecosystems. A known example of PHA production using microorganisms is to produce a PHA by feeding bacteria of the genus *Cupriavidus* with a carbon source such as a sugar, vegetable oil, or fatty acid and thus allowing the bacteria to accumulate the PHA in their cells (see Non Patent Literatures 2 and 3).

However, microbial material production requires the complicated steps of separating and collecting the microbial cells and the target product and could suffer the problem of high production cost. Improving the efficiency of separation and collection is a major challenge to be addressed for production cost reduction.

CITATION LIST

Non-Patent Literature

NPL 1: Anderson A. J. et al., *Int. J. Biol. Macromol.*, 12, 102-105 (1990)

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NPL 2: Sato S. et al., *J. Biosci. Bioeng.*, 120 (3), 246-251 (2015)

NPL 3: Insomphun C. et al., *Metab. Eng.*, 27, 38-45 (2015)

SUMMARY OF INVENTION

Technical Problem

A PHA is accumulated in microbial cells. To use the PHA accumulated in the microbial cells as a biodegradable plastic, it is necessary first to separate and collect the microbial cells from the culture fluid. The separation and collection of the microbial cells can be conducted by means such as a centrifuge or separation membrane, and the ease and efficiency of the separation and collection depend on the size of the microbial cells. Specifically, a larger size of the microbial cells allows the separation and collection to be more easily and efficiently accomplished by means such as a centrifuge or separation membrane, leading to a lower production cost.

The microbial cells accumulating the PHA are broken to take PHA particles out of the cells, and the PHA particles are separated from other cellular components and collected. Techniques for the separation and collection of the PHA particles are broadly classified into a technique using an organic solvent system and a technique using an aqueous system. Since the use of an organic solvent causes high environmental load and involves high cost, the technique using an aqueous system is preferred from the industrial point of view. With the technique using an aqueous system, for example, the PHA particles contained in the broken cell fluid can be separated from the fluid by means such as a centrifuge or separation membrane. In this case, the efficiency of the separation and collection depends on the size of the PHA particles. Specifically, a larger size of the PHA particles accumulated in the microbial cells allows the separation and collection to be more easily accomplished by means such as a centrifuge or separation membrane, leading to a lower production cost.

In view of the above circumstances, the present invention aims to provide a transformed microorganism that accumulates a PHA and whose size can be large and a method of producing the PHA using the transformed microorganism.

Solution to Problem

As a result of intensive studies, the present inventors have found that when the expression of a particular one of genes expected to be involved in cell division, namely, a minC gene (e.g., a gene that encodes the amino acid sequence of SEQ ID NO: 1), a minD gene (e.g., a gene that encodes the amino acid sequence of SEQ ID NO: 2), and a minE gene (e.g., a gene that encodes the amino acid sequence of SEQ ID NO: 3), is enhanced or reduced, the size of microbial cells can be increased while ensuring an industrially desired level of PHA accumulation. Based on this finding, the inventors have arrived at the present invention.

That is, the present invention relates to a transformed microorganism having a polyhydroxyalkanoate synthase gene, wherein expression of a minD gene is enhanced. The present invention further relates to a transformed microorganism having a polyhydroxyalkanoate synthase gene, wherein expression of a minC gene and a minD gene is enhanced. In this transformed microorganism, expression of a minE gene may be enhanced or reduced. The above transformed microorganisms preferably belong to the genus *Cupriavidus* and are more preferably transformed *Cupria-*

vidus necator. The present invention further relates to a method of producing a polyhydroxyalkanoate, the method including the step of culturing any of the transformed microorganisms in the presence of a carbon source. The carbon source preferably contains an oil, a fatty acid, a sugar, or carbon dioxide. The polyhydroxyalkanoate is preferably a copolymer of two or more hydroxyalkanoates, more preferably a copolymer containing 3-hydroxyhexanoate as a monomer unit, and even more preferably a copolymer of 3-hydroxybutyrate and 3-hydroxyhexanoate.

Advantageous Effects of Invention

The present invention can provide a transformed microorganism that accumulates a PHA and whose size can be large and a method of producing the PHA using the transformed microorganism. In the present invention, since the size of microbial cells accumulating the PHA is large, the microbial cells can easily be separated and collected from a culture fluid, and the production cost can be reduced.

A preferred aspect of the present invention can provide a transformed microorganism whose size can be large and that is capable of accumulating large-size PHA particles and a method of producing a PHA using the transformed microorganism. In this aspect, not only are the separation and collection of microbial cells from a culture fluid easy, but also large-size PHA particles are accumulated in the microbial cells. Thus, the PHA can easily be collected separately from other cellular components after cell breakage, and the production cost can be reduced.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a microscope image of a cultured KNK-005 strain (Comparative Example 1), where the scale bar represents 10 μ m (the same applies to FIGS. 2 to 8).

FIG. 2 is a microscope image of a cultured minE gene-deleted strain (Comparative Example 2).

FIG. 3 is a microscope image of a cultured minC gene expression-enhanced strain (Comparative Example 3).

FIG. 4 is a microscope image of a cultured minD gene expression-enhanced and minE gene-deleted strain (Comparative Example 4).

FIG. 5 is a microscope image of a cultured minD gene expression-enhanced strain (Example 1).

FIG. 6 is a microscope image of a cultured minCD gene expression-enhanced strain (Example 2).

FIG. 7 is a microscope image of a cultured minCDE gene expression-enhanced strain (Example 3).

FIG. 8 is a microscope image of a cultured minCD gene expression-enhanced and minE gene-deleted strain (Example 4).

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of the present invention will be described in detail. A transformed microorganism according to the present invention is a transformed microorganism that has a PHA synthase gene and in which expression of a particular one of minC, minD, and minE genes is enhanced or reduced.

Microorganism

The transformed microorganism according to the present invention may be a microorganism having a PHA synthase gene and transformed to enhance the expression of the minD

gene. Alternatively, the transformed microorganism may be a microorganism having a PHA synthase gene and transformed to enhance the expression of the minC and minD genes. Alternatively, the transformed microorganism may be a microorganism having a PHA synthase gene and transformed to enhance the expression of the minC, minD, and minE genes or may be a microorganism having a PHA synthase gene and transformed to enhance the expression of the minC and minD genes and reduce the expression of the minE gene. It should be noted that a microorganism transformed to enhance the expression of the minD gene and reduce the expression of the minE gene and not having enhanced expression of the minC gene is not categorized as the transformed microorganism of the present invention.

The host of the transformed microorganism according to the present invention is not limited to a particular type, and may be any microorganism that has a PHA synthase gene. The host is preferably a bacterium having a minCD or minCDE gene. Examples of the bacterium include bacteria belonging to the genus *Ralstonia*, the genus *Cupriavidus*, the genus *Wautersia*, the genus *Aeromonas*, the genus *Escherichia*, the genus *Alcaligenes*, and the genus *Pseudomonas*. In view of safety and PHA productivity, bacteria belonging to the genus *Ralstonia*, the genus *Cupriavidus*, the genus *Aeromonas*, and the genus *Wautersia* are more preferred. Even more preferred are bacteria belonging to the genus *Cupriavidus* or the genus *Aeromonas*, and still even more preferred are microorganisms belonging to the genus *Cupriavidus*. Particularly preferred is *Cupriavidus necator*.

The host of the transformed microorganism according to the present invention may be a wild strain inherently having a PHA synthase gene, a mutant strain obtained by artificially mutating the wild strain, or a strain having a foreign PHA synthase gene introduced by a genetic engineering technique. The introduction of the foreign PHA synthase gene is not limited to being carried out by a particular method, and the introduction method can be selected from: a method in which the foreign gene is directly inserted onto the chromosome of the host or a gene on the chromosome is replaced by the foreign gene; a method in which the foreign gene is directly inserted onto the megaplasmid of the host or a gene on the megaplasmid is replaced by the foreign gene; and a method in which the foreign gene is placed on a vector such as a plasmid, phage, or phagemid and the vector with the gene is introduced into the host. Two or more of these methods may be used in combination. In view of the stability of the introduced gene, it is preferable to use the method in which the foreign gene is directly inserted onto the chromosome of the host or a gene on the chromosome is replaced by the foreign gene or the method in which the foreign gene is directly inserted onto the megaplasmid of the host or a gene on the megaplasmid is replaced by the foreign gene, and it is more preferable to use the method in which the foreign gene is directly inserted onto the chromosome of the host or a gene on the chromosome is replaced by the foreign gene.

PHA Synthase Gene

The PHA synthase gene is not limited to a particular type, and examples of the PHA synthase gene include PHA synthase genes derived from living organisms belonging to the genus *Ralstonia*, the genus *Cupriavidus*, the genus *Wautersia*, the genus *Alcaligenes*, the genus *Aeromonas*, the genus *Pseudomonas*, the genus *Norcardia*, and the genus *Chromobacterium*, and further include altered genes resulting from alteration of the mentioned PHA synthase genes.

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Such an altered gene may be a gene having a base sequence that encodes a PHA synthase in which one or more amino acid residues are deleted, added, inserted, or replaced. Examples of the altered gene include a gene having a base sequence that encodes a polypeptide represented by an amino acid sequence of any one of SEQ ID NOS: 4 to 8 and a gene having a base sequence that encodes a polypeptide having PHA synthase activity and represented by an amino acid sequence that is at least 85% homologous to the amino acid sequence of any one of SEQ ID NOS: 4 to 8. The sequence homology is preferably 90% or more, more preferably 95% or more, even more preferably 97% or more, and particularly preferably 99% or more.

PHA

The PHA produced by the transformed microorganism of the present invention is not limited to a particular type, and may be any PHA that can be produced by microorganisms. The PHA is preferably any one of the following polymers: a homopolymer of one monomer selected from 3-hydroxyalkanoates having 4 to 16 carbon atoms; a copolymer of one monomer selected from 3-hydroxyalkanoates having 4 to 16 carbon atoms and another hydroxyalkanoic acid (such as a 2-hydroxyalkanoic acid, 4-hydroxyalkanoic acid, 5-hydroxyalkanoic acid, or 6-hydroxyalkanoic acid having 4 to 16 carbon atoms); and a copolymer of two or more monomers selected from 3-hydroxyalkanoates having 4 to 16 carbon atoms. Examples of the PHA include, but are not limited to: P(3HB) which is a homopolymer of 3-hydroxybutyrate (abbreviated as 3HB); P(3HB-co-3HV) which is a copolymer of 3HB and 3-hydroxyvaleric acid (abbreviated as 3HV); P(3HB-co-3HH) (abbreviated as PHBH) which is a copolymer of 3HB and 3-hydroxyhexanoate (abbreviated as 3HH); P(3HB-co-4HB) which is a copolymer of 3HB and 4-hydroxybutyrate (abbreviated as 4HB); and PHA containing lactic acid (abbreviated as LA) as a constituent component (an example of this PHA is P(LA-co-3HB) which is a copolymer of 3HB and LA). Among these examples, PHBH is preferred in that this polymer has a wide range of applications. The type of the PHA to be produced can be appropriately selected according to the intended purpose and depending on the type of the PHA synthase gene possessed by or introduced into the microorganism used, the type of the metabolizing gene involved in synthesis of the PHA, and the culture conditions.

minC, minD, and minE Genes

Proteins MinC, MinD, and MinE encoded by the minC, minD, and minE genes are proteins that cooperate in bacteria to control cell division (MinCDE system). For example, it is known that in cells of *Escherichia coli*, the MinD forms a polymer in an ATP-dependent manner, further forms a complex with the MinC, and rapidly oscillates between the cell poles. The MinC serves to inhibit septum formation during cell division. The MinE is known to bind to the MinD competitively against the MinC, and serves to regulate septum formation so that the septum is formed only at the center of the cell.

The minC gene is a gene having a base sequence that encodes a polypeptide (UniProtKB ID Q0KF13) represented by the amino acid sequence of SEQ ID NO: 1 and a polypeptide represented by an amino acid sequence that is at least 85% homologous to the amino acid sequence of SEQ ID NO: 1. The sequence homology is preferably 90% or

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more, more preferably 95% or more, even more preferably 97% or more, and particularly preferably 99% or more.

The minD gene is a gene having a base sequence that encodes a polypeptide (UniProtKB ID Q0KF14) represented by the amino acid sequence of SEQ ID NO: 2 and a polypeptide represented by an amino acid sequence that is at least 85% homologous to the amino acid sequence of SEQ ID NO: 2. The sequence homology is preferably 90% or more, more preferably 95% or more, even more preferably 97% or more, and particularly preferably 99% or more.

The minE gene is a gene having a base sequence that encodes a polypeptide (UniProtKB ID Q0KF15) represented by the amino acid sequence of SEQ ID NO: 3 and a polypeptide represented by an amino acid sequence that is at least 85% homologous to the amino acid sequence of SEQ ID NO: 3. The sequence homology is preferably 90% or more, more preferably 95% or more, even more preferably 97% or more, and particularly preferably 99% or more.

Gene Expression Enhancement

In the present invention, enhanced gene expression means a state in which the amount of transcription of a target gene or the amount of expression of the polypeptide encoded by the target gene is increased as compared to that in a strain in which the expression of the target gene is not enhanced. The increase is not limited to a particular level, and it is sufficient that the amount of transcription of the target gene or the amount of expression of the polypeptide be more than that in the strain in which the expression of the target gene is not enhanced. The amount of transcription of the target gene or the amount of expression of the polypeptide is preferably 1.1 or more times, more preferably 1.2 or more times, even more preferably 1.5 or more times, and still even more preferably 2 or more times that in the strain in which the expression of the target gene is not enhanced.

In the present invention, the enhancement of the expression of the min genes is not limited to being carried out by a particular method, and the enhancement method can be selected from a method in which the target gene is introduced into the host and a method in which the amount of expression of the target gene inherently possessed by the host on the genome DNA is increased. Both of the two methods may be used in combination.

The introduction of the target gene into the host is not limited to being carried out by a particular method, and the introduction method can be selected from: a method in which the target gene is directly inserted onto the chromosome of the host or a gene on the chromosome is replaced by the target gene; a method in which the target gene is directly inserted onto the megaplasmid of the host or a gene on the megaplasmid is replaced by the target gene; and a method in which the target gene is placed on a vector such as a plasmid, phage, or phagemid and the vector with the gene is introduced into the host. Two or more of these methods may be used in combination.

In view of the stability of the introduced gene, it is preferable to use the method in which the target gene is directly inserted onto the chromosome of the host or a gene on the chromosome is replaced by the target gene or the method in which the target gene is directly inserted onto the megaplasmid of the host or a gene on the megaplasmid is replaced by the target gene, and it is more preferable to use the method in which the target gene is directly inserted onto the chromosome of the host or a gene on the chromosome is replaced by the target gene. For reliable expression of the introduced gene, it is preferable to introduce the target gene

in such a manner that the target gene is downstream of a “gene expression regulatory sequence” inherently possessed by the host or downstream of a foreign “gene expression regulatory sequence”. In the present invention, a “gene expression regulatory sequence” is a DNA sequence including a base sequence that controls the amount of transcription of the gene (an example of this base sequence is a promoter sequence) and/or a base sequence that regulates the amount of translation of a messenger RNA transcribed from the gene (an example of this base sequence is a Shine-Dalgarno sequence). The “gene expression regulatory sequence” used may be any suitable naturally-occurring base sequence or an artificially constructed or altered base sequence.

The increase of the amount of expression of the target gene inherently possessed by the host on the genome DNA is not limited to being achieved by a particular method, and exemplary methods include a method in which a “gene expression regulatory sequence” upstream of the target gene is altered, a method in which a foreign “gene expression regulatory sequence” is introduced upstream of the target gene, and a method in which the target gene and/or a base sequence neighboring the target gene is altered to increase the stability of the transcribed messenger RNA.

Examples of the promoter sequence or Shine-Dalgarno sequence included in the “gene expression regulatory sequence” include, but are not limited to, the base sequences of SEQ ID NOS: 9 to 15 and base sequences including any part of the base sequences of SEQ ID NOS: 9 to 15.

Replacement, deletion, insertion, and/or addition made to at least a part of the genome DNA can be accomplished using a method known to those skilled in the art. Typical methods include a method using a transposon and the mechanism of homologous recombination (Ohman et al., *J. Bacteriol.*, 162:1068-1074 (1985)) and a method based on site-specific integration caused by the mechanism of homologous recombination and on loss due to secondary homologous recombination (Noti et al., *Methods Enzymol.*, 154:197-217 (1987)). A method may also be used in which a *sacB* gene derived from *Bacillus subtilis* is allowed to coexist and in which a microorganism strain having lost a gene due to secondary homologous recombination is easily isolated as a sucrose-resistant strain (Schweizer, *Mol. Microbiol.*, 6:1195-1204 (1992) or Lenz et al., *J. Bacteriol.*, 176:4385-4393 (1994)). Another alternative method is to use a CRISPR/Cas9 system-based genome-editing technology for altering the target DNA (Y. Wang et al., *ACS Synth Biol.*, 2016, 5 (7):721-732). In the CRISPR/Cas9 system, the guide RNA (gRNA) has a sequence capable of binding to a part of the base sequence of the genome DNA to be altered, and serves to transport the Cas9 to the target.

The introduction of a vector into a cell is not limited to being carried out by a particular method, and exemplary methods include calcium chloride transformation, electroporation, polyethylene glycol transformation, and spheroplast transformation.

Gene Expression Reduction

In the present invention, “reduced gene expression” means a state in which the amount of transcription of a target gene or the amount of expression of the polypeptide encoded by the target gene is decreased as compared to that in a strain in which the expression of the target gene is not reduced. The decrease is not limited to a particular level, and it is sufficient that the amount of transcription of the target gene or the amount of expression of the polypeptide be less than that in the strain in which the expression of the target gene

is not reduced. The amount of transcription of the target gene or the amount of expression of the polypeptide is preferably 0.8 or less times, more preferably 0.5 or less times, even more preferably 0.3 or less times, and still even more preferably 0.2 or less times that in the strain in which the expression of the target gene is not reduced. The amount of transcription of the target gene or the amount of expression of the polypeptide encoded by the target gene may be zero. The gene expression can be considered to have been reduced also when the polypeptide encoded by the target gene cannot exhibit the original function for a reason such as alteration of the base sequence of the gene. In the case where the transformed microorganism of the present invention is a transformed microorganism with enhanced expression of the *minC* and *minD* genes, the expression of the target gene can be reduced by using a drug or protein that inhibits the function of the corresponding polypeptide.

In the present invention, the reduction of the gene expression is not limited to being achieved by a particular method, and exemplary methods include: a method in which a part or the entire length of the target gene is deleted; a method in which the “gene expression regulatory sequence” involved in the expression of the target gene is altered; and a method in which the target gene and/or a base sequence neighboring the target gene is altered to decrease the stability of the transcribed messenger RNA. The base sequence alteration is not limited to being carried out by a particular method, and can be accomplished through replacement, deletion, insertion, and/or addition made to at least a part of the target gene and/or the neighboring base sequence. The replacement, deletion, insertion, and/or addition can be made by a method known to those skilled in the art. In the case where the transformed microorganism of the present invention is a transformed microorganism with enhanced expression of the *minC* and *minD* genes, an antisense RNA, RNA interference (RNAi), or CRISPR interference (CRISPRi) may be used to reduce the expression of the target gene without altering the target gene and/or the neighboring base sequence.

Culturing the transformed microorganism of the present invention allows the microorganism to accumulate a PHA therein. The culture of the transformed microorganism of the present invention can be conducted according to a common microbial culture method, and it is sufficient that the transformed microorganism be cultured in a culture medium containing a suitable carbon source. There are no particular limitations on the composition of the culture medium, the method of adding the carbon source, the scale of the culture, the conditions of aeration and stirring, the culture temperature, and the culture time. It is preferable to add the carbon source continuously or intermittently to the culture medium.

The carbon source used for the culture may be any carbon source that can be assimilated by the transformed microorganism of the present invention. Examples of the carbon source include, but are not limited to: sugars such as glucose, fructose, and sucrose; palm and palm kernel oils (including palm olein, palm double olein, and palm kernel olein which are low-melting fractions obtained through fractionation of palm oil and palm kernel oil); oils such as corn oil, coconut oil, olive oil, soybean oil, rapeseed oil, and Jatropha oil; fractions of these oils; by-products formed during refining of these oils; fatty acids such as lauric acid, oleic acid, stearic acid, palmitic acid, and myristic acid; derivatives of these fatty acids; and glycerol. In the case where the transformed microorganism of the present invention can assimilate gases such as carbon dioxide, carbon monoxide, and methane or alcohols such as methanol and ethanol, any of these gases or alcohols can be used as the carbon source.

In the PHA production of the present invention, it is preferable to culture the microorganism using a culture medium containing the carbon source and other nutrient sources including a nitrogen source, an inorganic salt, and another organic nutrient source. Examples of the nitrogen source include, but are not limited to: ammonia; ammonium salts such as ammonium chloride, ammonium sulfate, and ammonium phosphate; peptone; meat extracts; and yeast extracts. Examples of the inorganic salt include potassium dihydrogen phosphate, sodium dihydrogen phosphate, magnesium phosphate, magnesium sulfate, and sodium chloride. Examples of the other organic nutrient source include: amino acids such as glycine, alanine, serine, threonine, and proline; and vitamins such as vitamin B1, vitamin B12, and vitamin C.

After the microorganism is cultured for an adequate time to allow the microorganism to accumulate a PHA therein, the PHA is collected from the microorganism using a known method. The PHA collection is not limited to being carried out by a particular method. For example, the PHA can be collected by a method consisting of: after the culture, separating the microorganism from the culture fluid by means such as a centrifuge or separation membrane; drying the separated microorganism; extracting the PHA from the dried microorganism using an organic solvent such as chloroform; removing cellular components from the PHA-containing organic solvent solution by a process such as filtration; adding a poor solvent such as methanol or hexane to the filtrate to precipitate the PHA; removing the supernatant by a process such as filtration or centrifugation; and drying the precipitated PHA. Alternatively, the PHA may be collected by dissolving cellular components other than the PHA in water with the aid of a surfactant, an alkali, or an enzyme, then separating the PHA particles from the aqueous phase by a process such as filtration or centrifugation, and drying the separated PHA particles.

In the present invention, large-size microbial cells accumulating PHA can be obtained, and the microbial cells can be separated from the culture fluid easily and efficiently thanks to their large size. Large-size PHA particles producible according to a preferred aspect of the present invention are preferred because such PHA particles are easy to separate and collect using an aqueous system as described above.

EXAMPLES

Hereinafter, the present invention will be described more specifically using examples. The present invention is not limited to the examples. The overall genetic manipulation can be carried out, for example, in a manner as taught in Molecular Cloning (Cold Spring Harbor Laboratory Press (1989)). The enzymes and cloning hosts used in the gene manipulation can be purchased from market suppliers and used according to the instructions given by the suppliers. The enzymes are not limited to particular types and may be any enzymes that can be used for gene manipulation.

(Production Example 1) Preparation of minE Gene-Deleted Strain

First, a gene deletion plasmid was prepared. The preparation was done as follows. PCR using a synthetic oligo DNA was carried out to obtain a DNA fragment (SEQ ID NO: 16) having base sequences upstream and downstream of the minE structural gene. The DNA fragment was digested by a restriction enzyme *Swa*I, and the resulting DNA fragment was joined by a DNA ligase (Ligation High,

manufactured by Toyobo Co., Ltd.) to a vector pNS2X-sacB which is described in Japanese Laid-Open Patent Application Publication No. 2007-259708 and which was also digested by *Swa*I. Thus, a gene deletion plasmid vector pNS2X-sacB+minEUD having base sequences upstream and downstream of the minE structural gene was prepared.

Subsequently, a minE gene-deleted strain was prepared using the gene deletion plasmid vector pNS2X-sacB+minEUD as follows. An *Escherichia coli* S17-1 strain (ATCC 47055) was transformed with the gene deletion plasmid vector pNS2X-sacB+minEUD, and the resulting transformed microorganism was cocultured with a KNK-005 strain on Nutrient Agar (manufactured by Difco Laboratories) to effect conjugal transfer. The KNK-005 strain is a transformed strain produced by introducing an *Aeromonas caviae*-derived PHA synthase gene (a gene that encodes a PHA synthase that has the amino acid sequence of SEQ ID NO: 6) onto the chromosome of a *Cupriavidus necator* H16 strain, and can be prepared according to the method described in U.S. Pat. No. 7,384,766.

The culture fluid obtained as above was inoculated into a Simmons agar medium (2 g/L sodium citrate, 5 g/L sodium chloride, 0.2 g/L magnesium sulfate heptahydrate, 1 g/L ammonium dihydrogen phosphate, 1 g/L potassium dihydrogen phosphate, 15 g/L agar, pH=6.8) containing 250 mg/L of kanamycin, and a strain grown on the agar medium was selectively collected. Thus, a strain having the plasmid integrated into the chromosome of the KNK-005 strain was obtained. The obtained strain was cultured on Nutrient Broth (manufactured by Difco Laboratories) for two generations, after which the culture broth was diluted and applied onto Nutrient Agar containing 15% sucrose. A strain grown on Nutrient Agar was obtained as a strain having lost the plasmid. PCR and analysis using a DNA sequencer were further carried out to isolate one strain from which the start to stop codons of the minE structural gene on the chromosome were deleted. In this manner, a minE gene-deleted strain was obtained.

(Production Example 2) Preparation of minC Gene Expression-Enhanced Strain

First, a minC gene expression plasmid pCUP2-PA-minC was prepared. The preparation was done as follows.

PCR using a synthetic oligo DNA was carried out to obtain a DNA fragment (SEQ ID NO: 17) having a promoter sequence and a minC gene sequence. The DNA fragment was digested by restriction enzymes *Mun*I and *Spe*I, and the resulting DNA fragment was joined to a plasmid vector pCUP2 which is described in WO 2007/049716 and which was cleaved by *Mun*I and *Spe*I. Thus, the minC gene expression plasmid pCUP2-PA-minC was obtained.

Subsequently, the minC gene expression plasmid pCUP2-PA-minC was introduced into the KNK-005 strain to obtain a minC gene expression-enhanced strain. The introduction of the plasmid vector into the cells was accomplished by electroporation. The gene introduction device used was Gene Pulser manufactured by Bio-Rad Laboratories, Inc., and the cuvette used was a 0.2-cm-gap cuvette also manufactured by Bio-Rad Laboratories, Inc. The cuvette was charged with 400 μ l of competent cells and 20 μ l of an expression vector and set on the pulse device, by which electric pulse was applied to the contents of the cuvette at a capacitance of 25 μ F, a voltage of 1.5 kV, and a resistance value of 800 Ω . After the pulse application, the fluid in the cuvette was subjected to shake culture on Nutrient Broth (manufactured by Difco Laboratories) at 30° C. for 3 hours

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and then to culture on a selection plate (Nutrient Agar manufactured by Difco Laboratories, containing 100 mg/L kanamycin) at 30° C. for 2 days. The minC gene expression-enhanced strain thus grown was collected.

(Production Example 3) Preparation of minD Gene Expression-Enhanced Strain

First, a minD gene expression plasmid pCUP2-PA-minD was prepared. The preparation was done as follows.

PCR using a synthetic oligo DNA was carried out to obtain a DNA fragment (SEQ ID NO: 18) having a promoter sequence and a minD gene sequence. The DNA fragment was digested by restriction enzymes MunI and SpeI, and the resulting DNA fragment was joined to a plasmid vector pCUP2 which is described in WO 2007/049716 and which was cleaved by MunI and SpeI. Thus, the minD gene expression plasmid pCUP2-PA-minD was obtained.

Subsequently, the minD gene expression plasmid pCUP2-PA-minD was introduced into the KNK-005 strain in the same manner as the plasmid pCUP2-PA-minC was introduced in Production Example 2. Thus, a minD gene expression-enhanced strain was obtained.

(Production Example 4) Preparation of minCD Gene Expression-Enhanced Strain

First, a minCD gene expression plasmid pCUP2-PA-minCD was prepared. The preparation was done as follows.

PCR using a synthetic oligo DNA was carried out to obtain a DNA fragment (SEQ ID NO: 19) having a promoter sequence and a minCD gene sequence. The DNA fragment was digested by restriction enzymes MunI and SpeI, and the resulting DNA fragment was joined to a plasmid vector pCUP2 which is described in WO 2007/049716 and which was cleaved by MunI and SpeI. Thus, the minCD gene expression plasmid pCUP2-PA-minCD was obtained.

Subsequently, the minCD gene expression plasmid pCUP2-PA-minCD was introduced into the KNK-005 strain in the same manner as the plasmid pCUP2-PA-minC was introduced in Production Example 2. Thus, a minCD gene expression-enhanced strain was obtained.

(Production Example 5) Preparation of minD Gene Expression-Enhanced and minE Gene-Deleted Strain

The minD gene expression plasmid pCUP2-PA-minD prepared in Production Example 3 was introduced into the minE gene-deleted strain prepared in Production Example 1 in the same manner as the plasmid pCUP2-PA-minC was introduced into the KNK-005 strain in Production Example 2. Thus, a minD gene expression-enhanced and minE gene-deleted strain was obtained.

(Production Example 6) Preparation of minCDE Gene Expression-Enhanced Expression

First, a minCDE gene expression plasmid pCUP2-PA-minCDE was prepared. The preparation was done as follows.

PCR using a synthetic oligo DNA was carried out to obtain a DNA fragment (SEQ ID NO: 20) having a promoter sequence and a minCDE gene sequence. The DNA fragment was digested by restriction enzymes MunI and SpeI, and the resulting DNA fragment was joined to a plasmid vector pCUP2 which is described in WO 2007/049716 and which

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was cleaved by MunI and SpeI. Thus, the minCDE gene expression plasmid pCUP2-PA-minCDE was obtained.

Subsequently, the minCDE gene expression plasmid pCUP2-PA-minCDE was introduced into the KNK-005 strain in the same manner as the plasmid pCUP2-PA-minC was introduced in Production Example 2. Thus, a minCDE gene expression-enhanced strain was obtained.

(Production Example 7) Preparation of minCD Gene Expression-Enhanced and minE Gene-Deleted Strain

The minCD gene expression plasmid pCUP2-PA-minCD prepared in Production Example 4 was introduced into the minE gene-deleted strain prepared in Production Example 1 in the same manner as the plasmid pCUP2-PA-minC was introduced into the KNK-005 strain in Production Example 2. Thus, a minCD gene expression-enhanced and minE gene-deleted strain was obtained.

(Comparative Example 1) PHA Production by KNK-005 Strain

Culture examination using the KNK-005 strain was conducted under the conditions described below.

Culture Media

The seed culture medium was composed of 1 w/v % Meat-extract, 1 w/v % Bacto-Tryptone, 0.2 w/v % Yeast-extract, 0.9 w/v % Na₂HPO₄·12H₂O, and 0.15 w/v % KH₂PO₄ (pH=6.8). The preculture medium was composed of 1.1 w/v % Na₂HPO₄·12H₂O, 0.19 w/v % KH₂PO₄, 1.29 w/v % (NH₄)₂SO₄, 0.1 w/v % MgSO₄·7H₂O, 2.5 w/v % palm olein oil, and 0.5 v/v % trace metal salt solution (solution of 1.6 w/v % FeCl₃·6H₂O, 1 w/v % CaCl₂·2H₂O, 0.02 w/v % CoCl₂·6H₂O, 0.016 w/v % CuSO₄·5H₂O, and 0.012 w/v % NiCl₂·6H₂O in 0.1N hydrochloric acid). Palm olein oil was added as a carbon source in a concentration of 10 g/L at one time. The PHA production culture medium was composed of 0.385 w/v % Na₂HPO₄·12H₂O, 0.067 w/v % KH₂PO₄, 0.291 w/v % (NH₄)₂SO₄, 0.1 w/v % MgSO₄·7H₂O, and 0.5 v/v % trace metal salt solution (solution of 1.6 w/v % FeCl₃·6H₂O, 1 w/v % CaCl₂·2H₂O, 0.02 w/v % CoCl₂·6H₂O, 0.016 w/v % CuSO₄·5H₂O, and 0.012 w/v % NiCl₂·6H₂O in 0.1N hydrochloric acid).

Method of Measuring Accumulated PHA Percentage

The accumulated PHA percentage was measured as follows. The microorganism was collected from the culture fluid by centrifugation. The collected microorganism was washed with ethanol and freeze-dried to give a dried microorganism, the weight of which was measured. To 1 g of the dried microorganism was added 100 ml of chloroform, and the microorganism in chloroform was stirred at room temperature for a day to extract a PHA from the microorganism. The residual microorganism was removed by filtration, and the filtrate was concentrated using an evaporator to a total volume of 30 ml. To the concentrate was slowly added 90 ml of hexane, and the mixture was left for 1 hour under gentle stirring. The PHA precipitated was collected by filtration and vacuum-dried at 50° C. for 3 hours. The weight of the dried PHA was measured, and the percentage of the accumulated PHA to the dried microorganism was calculated.

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Method of Measuring Cell Size

The cell size was measured as follows. After the culture, the culture fluid was treated at 65° C. for 60 minutes to inactivate the microbial cells. The treated fluid was analyzed with a laser diffraction-scattering particle size distribution analyzer (Microtrac MT3300EXII) to measure the mean volume diameter (MV) of the cells. The measurement was conducted using standard settings (Permeability: Transparent, Particle refractive index: 1.81, Particle shape: Non-spherical, Solvent refractive index: 1.333).

Method of Measuring PHA Particle Size

The PHA particle size was measured as follows. After the culture, the culture fluid was treated at 65° C. for 60 minutes to inactivate the microbial cells. The culture fluid was diluted to 150 times the original volume with a 3.3 w/v % aqueous solution of sodium dodecyl sulfate, and subjected to ultrasonic disintegration to obtain a liquid containing the extracted PHA. The ultrasonic disintegration was conducted using Ultrasonic Homogenizer UH-600 manufactured by SMT Co., Ltd. and consisted of four repetitions of ultrasonic stirring performed at a maximum output for 40 seconds. The resulting liquid containing the extracted PHA was analyzed with a laser diffraction-scattering particle size distribution analyzer (Microtrac MT3300EXII) to measure the mean volume diameter (MV) of the PHA particles. The measurement was conducted using standard settings (Permeability: Transparent, Particle refractive index: 1.81, Particle shape: Non-spherical, Solvent refractive index: 1.333).

Microscopic Observation of Cells

Microscopic observation of the cells was conducted as follows. After the culture, the culture fluid was diluted as appropriate. The dilution was placed and dried on a glass slide, and then the cells were stained with fuchsin. The stained cells were observed with an optical microscope.

PHA Production Culture

PHA production culture was performed as follows. First, a glycerol stock (50 µl) of the KNK-005 strain was inoculated into the seed culture medium (10 ml) and cultured for 24 hours to accomplish seed culture. Subsequently, the seed culture fluid was inoculated at a concentration of 1.0 v/v % into a 3 L jar fermenter (MDL-300, manufactured by B. E. Marubishi Co., Ltd.) containing 1.8 L of the preculture medium. The fermenter was operated at a culture temperature of 33° C., a stirring speed of 500 rpm, and an aeration of 1.8 L/min, and the preculture was conducted for 28 hours during which the pH was controlled between 6.7 and 6.8. For the pH control, a 14% aqueous solution of ammonium hydroxide was used.

Next, the preculture fluid was inoculated at a concentration of 5.0 v/v % into a 5 L jar fermenter (MDS-U50, manufactured by B. E. Marubishi Co., Ltd.) containing 2.5 L of the PHA production culture medium. The fermenter was operated at a culture temperature of 33° C., a stirring speed of 420 rpm, and an aeration of 2.1 L/min, and the pH was controlled between 6.7 and 6.8. For the pH control, a 25% aqueous solution of ammonium hydroxide was used. The carbon source was added intermittently. Palm olein oil was used as the carbon source. The culture was continued until the accumulated PHA percentage reached around 90%. The accumulated PHA percentage, the cell size, and the PHA

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particle size were measured as previously described. The results are listed in Table 1. An image taken by the microscopic cell observation conducted as previously described is shown in FIG. 1.

(Comparative Example 2) PHA Production by minE Gene-Deleted Strain

Culture examination using the minE gene-deleted strain was conducted under the same conditions as the culture examination in Comparative Example 1. The measurement results of the accumulated PHA percentage, the cell size, and the PHA particle size are listed in Table 1. An image taken by the microscopic cell observation conducted as previously described is shown in FIG. 2.

The results of the culture examination revealed that the cell size of the minE gene-deleted strain was little different from that of the KNK-005 strain which was a parent strain.

(Comparative Example 3) PHA Production by minC Gene Expression-Enhanced Strain

Culture examination using the minC gene expression-enhanced strain was conducted under the same conditions as the culture examination in Comparative Example 1. The measurement results of the accumulated PHA percentage, the cell size, and the PHA particle size are listed in Table 1. An image taken by the microscopic cell observation conducted as previously described is shown in FIG. 3.

The results of the culture examination revealed that the cell size of the minC gene expression-enhanced strain was smaller than that of the KNK-005 strain which was a parent strain. Additionally, the PHA productivity of the minC gene expression-enhanced strain was considerably low, and the accumulated PHA percentage was only 83% despite the culture time being longer than in Comparative Example 1.

(Comparative Example 4) PHA Production by minD Gene Expression-Enhanced and minE Gene-Deleted Strain

Culture examination using the minD gene expression-enhanced and minE gene-deleted strain was conducted under the same conditions as the culture examination in Comparative Example 1. The measurement results of the accumulated PHA percentage, the cell size, and the PHA particle size are listed in Table 1. An image taken by the microscopic cell observation conducted as previously described is shown in FIG. 4.

The results of the culture examination revealed that the cell size of the minD gene expression-enhanced and minE gene-deleted strain was little different from that of the KNK-005 strain which was a parent strain.

(Example 1) PHA Production by minD Gene Expression-Enhanced Strain

Culture examination using the minD gene expression-enhanced strain was conducted under the same conditions as the culture examination in Comparative Example 1. The measurement results of the accumulated PHA percentage, the cell size, and the PHA particle size are listed in Table 1. An image taken by the microscopic cell observation conducted as previously described is shown in FIG. 5.

The results of the culture examination revealed that the cell size of the minD gene expression-enhanced strain was more than 10% above that of the KNK-005 strain which was

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a parent strain. Additionally, the PHA productivity was comparable to that of the KNK-005 strain.

(Example 2) PHA Production by minCD Gene Expression-Enhanced Strain

Culture examination using the minCD gene expression-enhanced strain was conducted under the same conditions as the culture examination in Comparative Example 1. The measurement results of the accumulated PHA percentage, the cell size, and the PHA particle size are listed in Table 1. An image taken by the microscopic cell observation conducted as previously described is shown in FIG. 6.

The results of the culture examination revealed that the cell size of the minCD gene expression-enhanced strain was more than 15% above that of the KNK-005 strain which was a parent strain. Additionally, the PHA productivity was comparable to that of the KNK-005 strain. The particle size of the PHA produced by the minCD gene expression-enhanced strain was greater than the particle size of the PHA produced by the KNK-005 strain.

(Example 3) PHA Production by minCDE Gene Expression-Enhanced Strain

Culture examination using the minCDE gene expression-enhanced strain was conducted under the same conditions as the culture examination in Comparative Example 1. The measurement results of the accumulated PHA percentage, the cell size, and the PHA particle size are listed in Table 1. An image taken by the microscopic cell observation conducted as previously described is shown in FIG. 7.

The results of the culture examination revealed that the cell size of the minCDE gene expression-enhanced strain was more than 20% above that of the KNK-005 strain which was a parent strain. Additionally, the particle size of the PHA produced by the minCDE gene expression-enhanced strain was greater than the particle size of the PHA produced by the KNK-005 strain.

(Example 4) PHA Production by minCD Gene Expression-Enhanced and minE Gene-Deleted Strain

Culture examination using the minCD gene expression-enhanced and minE gene-deleted strain was conducted

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under the same conditions as the culture examination in Comparative Example 1. The measurement results of the accumulated PHA percentage, the cell size, and the PHA particle size are listed in Table 1. An image taken by the microscopic cell observation conducted as previously described is shown in FIG. 8.

The results of the culture examination revealed that the cell size of the minCD gene expression-enhanced and minE gene-deleted strain was more than 55% or more above that of the KNK-005 strain which was a parent strain. Additionally, the PHA productivity was almost comparable to that of the KNK-005 strain.

The PHA produced in the culture examinations in Comparative Examples and Examples was found to be PHBH by HPLC analysis.

TABLE 1

	Strain	Percentage of accumulated PHA to dried microorganism (%)	Cell size (μm)	PHA particle size (μm)
Comp. Example 1	KNK-005 strain	90	1.89	1.74
Comp. Example 2	minE gene-deleted strain	90	1.93	1.72
Comp. Example 3	minC gene expression-enhanced strain	83	1.58	1.34
Comp. Example 4	minD gene expression-enhanced and minE gene-deleted strain	88	1.94	1.71
Example 1	minD gene expression-enhanced strain	90	2.09	1.78
Example 2	minCD gene expression-enhanced strain	90	2.24	1.94
Example 3	minCDE gene expression-enhanced strain	90	2.34	1.87
Example 4	minCD gene expression-enhanced and minE gene-deleted strain	87	3.00	1.73

SEQUENCE LISTING

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 Val Ile Ala Glu Gly Asn Ile His Ile Tyr Ala Pro Leu Arg Gly Arg
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 Cys Met Glu Pro Glu Leu Ile Ser Ile Ala Gly Ile Tyr Arg Thr Ala
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 35 40 45
 Ile Met Gly Cys Glu Arg Arg Val Val Tyr Asp Leu Ile Asn Val Val
 50 55 60
 Gln Gly Glu Ala Asn Leu Arg Gln Ala Leu Ile Lys Asp Lys Lys Cys
 65 70 75 80
 Glu Asn Leu Phe Ile Leu Pro Ala Ser Gln Thr Arg Asp Lys Asp Ala
 85 90 95
 Leu Thr Arg Glu Gly Val Glu Lys Val Ile Asn Gly Leu Ile Glu Met
 100 105 110
 Asp Phe Glu Phe Ile Ile Cys Asp Ser Pro Ala Gly Ile Glu Ser Gly
 115 120 125
 Ala Leu Met Ala Met Tyr Phe Ala Asp Glu Ala Leu Ile Val Thr Asn
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 Pro Glu Val Ser Ser Val Arg Asp Ser Asp Arg Ile Leu Gly Ile Leu
 145 150 155 160
 Ala Ser Lys Thr Lys Arg Ala Ser Glu Gly Gly Asp Pro Ile Lys Glu
 165 170 175

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His Leu Leu Ile Thr Arg Tyr Asn Pro Lys Arg Val His Gly Gly Glu
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Gly Val Val Pro Glu Ser Glu Ala Val Leu His Ala Ser Asn Gln Gly
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Thr Pro Ala Ile His Leu Glu Gly Ser Asp Val Ala Asp Ala Tyr Gly
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35 40 45

Ala Ser Gly Ile Pro Gly Leu Asp Ala Leu Ala Gly Val Lys Ile Ala
50 55 60

Pro Ala Gln Leu Gly Asp Ile Gln Gln Arg Tyr Met Lys Asp Phe Ser
65 70 75 80

Ala Leu Trp Gln Ala Met Ala Glu Gly Lys Ala Glu Ala Thr Gly Pro
85 90 95

Leu His Asp Arg Arg Phe Ala Gly Asp Ala Trp Arg Thr Asn Leu Pro
100 105 110

Tyr Arg Phe Ala Ala Ala Phe Tyr Leu Leu Asn Ala Arg Ala Leu Thr
115 120 125

Glu Leu Ala Asp Ala Val Glu Ala Asp Ala Lys Thr Arg Gln Arg Ile
130 135 140

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			165						170					175	
Glu	Ser	Leu	Arg	Ala	Gly	Val	Arg	Asn	Met	Met	Glu	Asp	Leu	Thr	Arg
			180					185					190		
Gly	Lys	Ile	Ser	Gln	Thr	Asp	Glu	Ser	Ala	Phe	Glu	Val	Gly	Arg	Asn
	195						200					205			
Val	Ala	Val	Thr	Glu	Gly	Ala	Val	Val	Phe	Glu	Asn	Glu	Tyr	Phe	Gln
	210					215					220				
Leu	Leu	Gln	Tyr	Lys	Pro	Leu	Thr	Asp	Lys	Val	His	Ala	Arg	Pro	Leu
225					230					235					240
Leu	Met	Val	Pro	Pro	Cys	Ile	Asn	Lys	Tyr	Tyr	Ile	Leu	Asp	Leu	Gln
			245						250					255	
Pro	Glu	Ser	Ser	Leu	Val	Arg	His	Val	Val	Glu	Gln	Gly	His	Thr	Val
			260					265					270		
Phe	Leu	Val	Ser	Trp	Arg	Asn	Pro	Asp	Ala	Ser	Met	Ala	Gly	Ser	Thr
	275						280					285			
Trp	Asp	Asp	Tyr	Ile	Glu	His	Ala	Ala	Ile	Arg	Ala	Ile	Glu	Val	Ala
	290					295				300					
Arg	Asp	Ile	Ser	Gly	Gln	Asp	Lys	Ile	Asn	Val	Leu	Gly	Phe	Cys	Val
305					310					315					320
Gly	Gly	Thr	Ile	Val	Ser	Thr	Ala	Leu	Ala	Val	Leu	Ala	Ala	Arg	Gly
			325						330					335	
Glu	His	Pro	Ala	Ala	Ser	Val	Thr	Leu	Leu	Thr	Thr	Leu	Leu	Asp	Phe
			340					345				350			
Ala	Asp	Thr	Gly	Ile	Leu	Asp	Val	Phe	Val	Asp	Glu	Gly	His	Val	Gln
	355						360					365			
Leu	Arg	Glu	Ala	Thr	Leu	Gly	Gly	Gly	Ala	Gly	Ala	Pro	Cys	Ala	Leu
	370					375					380				
Leu	Arg	Gly	Leu	Glu	Leu	Ala	Asn	Thr	Phe	Ser	Phe	Leu	Arg	Pro	Asn
385					390					395					400
Asp	Leu	Val	Trp	Asn	Tyr	Val	Val	Asp	Asn	Tyr	Leu	Lys	Gly	Asn	Thr
			405						410					415	
Pro	Val	Pro	Phe	Asp	Leu	Leu	Phe	Trp	Asn	Gly	Asp	Ala	Thr	Asn	Leu
			420					425					430		
Pro	Gly	Pro	Trp	Tyr	Cys	Trp	Tyr	Leu	Arg	His	Thr	Tyr	Leu	Gln	Asn
	435					440						445			
Glu	Leu	Lys	Val	Pro	Gly	Lys	Leu	Thr	Val	Cys	Gly	Val	Pro	Val	Asp
	450					455					460				
Leu	Ala	Ser	Ile	Asp	Val	Pro	Thr	Tyr	Ile	Tyr	Gly	Ser	Arg	Glu	Asp
465					470					475					480
His	Ile	Val	Pro	Trp	Thr	Ala	Ala	Tyr	Ala	Ser	Thr	Ala	Leu	Leu	Ala
			485						490					495	
Asn	Lys	Leu	Arg	Phe	Val	Leu	Gly	Ala	Ser	Gly	His	Ile	Ala	Gly	Val
		500						505					510		
Ile	Asn	Pro	Pro	Ala	Lys	Asn	Lys	Arg	Ser	His	Trp	Thr	Asn	Asp	Ala
	515						520					525			
Leu	Pro	Glu	Ser	Pro	Gln	Gln	Trp	Leu	Ala	Gly	Ala	Ile	Glu	His	His
	530					535					540				
Gly	Ser	Trp	Trp	Pro	Asp	Trp	Thr	Ala	Trp	Leu	Ala	Gly	Gln	Ala	Gly
545					550					555					560
Ala	Lys	Arg	Ala	Ala	Pro	Ala	Asn	Tyr	Gly	Asn	Ala	Arg	Tyr	Arg	Ala

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565	570	575
Ile Glu Pro Ala Pro Gly Arg Tyr Val Lys Ala Lys Ala		
580	585	
 <210> SEQ ID NO 5		
<211> LENGTH: 594		
<212> TYPE: PRT		
<213> ORGANISM: Aeromonas caviae		
 <400> SEQUENCE: 5		
Met Ser Gln Pro Ser Tyr Gly Pro Leu Phe Glu Ala Leu Ala His Tyr		
1	5	10
Asn Asp Lys Leu Leu Ala Met Ala Lys Ala Gln Thr Glu Arg Thr Ala		
20	25	30
Gln Ala Leu Leu Gln Thr Asn Leu Asp Asp Leu Gly Gln Val Leu Glu		
35	40	45
Gln Gly Ser Gln Gln Pro Trp Gln Leu Ile Gln Ala Gln Met Asn Trp		
50	55	60
Trp Gln Asp Gln Leu Lys Leu Met Gln His Thr Leu Leu Lys Ser Ala		
65	70	75
Gly Gln Pro Ser Glu Pro Val Ile Thr Pro Glu Arg Ser Asp Arg Arg		
85	90	95
Phe Lys Ala Glu Ala Trp Ser Glu Gln Pro Ile Tyr Asp Tyr Leu Lys		
100	105	110
Gln Ser Tyr Leu Leu Thr Ala Arg His Leu Leu Ala Ser Val Asp Ala		
115	120	125
Leu Glu Gly Val Pro Gln Lys Ser Arg Glu Arg Leu Arg Phe Phe Thr		
130	135	140
Arg Gln Tyr Val Asn Ala Met Ala Pro Ser Asn Phe Leu Ala Thr Asn		
145	150	155
Pro Glu Leu Leu Lys Leu Thr Leu Glu Ser Asp Gly Gln Asn Leu Val		
165	170	175
Arg Gly Leu Ala Leu Leu Ala Glu Asp Leu Glu Arg Ser Ala Asp Gln		
180	185	190
Leu Asn Ile Arg Leu Thr Asp Glu Ser Ala Phe Glu Leu Gly Arg Asp		
195	200	205
Leu Ala Leu Thr Pro Gly Arg Val Val Gln Arg Thr Glu Leu Tyr Glu		
210	215	220
Leu Ile Gln Tyr Ser Pro Thr Thr Glu Thr Val Gly Lys Thr Pro Val		
225	230	235
Leu Ile Val Pro Pro Phe Ile Asn Lys Tyr Tyr Ile Met Asp Met Arg		
245	250	255
Pro Gln Asn Ser Leu Val Ala Trp Leu Val Ala Gln Gly Gln Thr Val		
260	265	270
Phe Met Ile Ser Trp Arg Asn Pro Gly Val Ala Gln Ala Gln Ile Asp		
275	280	285
Leu Asp Asp Tyr Val Val Asp Gly Val Ile Ala Ala Leu Asp Gly Val		
290	295	300
Glu Ala Ala Thr Gly Glu Arg Glu Val His Gly Ile Gly Tyr Cys Ile		
305	310	315
Gly Gly Thr Ala Leu Ser Leu Ala Met Gly Trp Leu Ala Ala Arg Arg		
325	330	335
Gln Lys Gln Arg Val Arg Thr Ala Thr Leu Phe Thr Thr Leu Leu Asp		
340	345	350

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Phe Ser Gln Pro Gly Glu Leu Gly Ile Phe Ile His Glu Pro Ile Ile
   355                               360                               365

Ala Ala Leu Glu Ala Gln Asn Glu Ala Lys Gly Ile Met Asp Gly Arg
   370                               375                               380

Gln Leu Ala Val Ser Phe Ser Leu Leu Arg Glu Asn Ser Leu Tyr Trp
   385                               390                               395                               400

Asn Tyr Tyr Ile Asp Ser Tyr Leu Lys Gly Gln Ser Pro Val Ala Phe
   405                               410                               415

Asp Leu Leu His Trp Asn Ser Asp Ser Thr Asn Val Ala Gly Lys Thr
   420                               425                               430

His Asn Ser Leu Leu Arg Arg Leu Tyr Leu Glu Asn Gln Leu Val Lys
   435                               440                               445

Gly Glu Leu Lys Ile Arg Asn Thr Arg Ile Asp Leu Gly Lys Val Lys
   450                               455                               460

Thr Pro Val Leu Leu Val Ser Ala Val Asp Asp His Ile Ala Leu Trp
   465                               470                               475                               480

Gln Gly Thr Trp Gln Gly Met Lys Leu Phe Gly Gly Glu Gln Arg Phe
   485                               490                               495

Leu Leu Ala Glu Ser Gly His Ile Ala Gly Ile Ile Asn Pro Pro Ala
   500                               505                               510

Ala Asn Lys Tyr Gly Phe Trp His Asn Gly Ala Glu Ala Glu Ser Pro
   515                               520                               525

Glu Ser Trp Leu Ala Gly Ala Thr His Gln Gly Gly Ser Trp Trp Pro
   530                               535                               540

Glu Met Met Gly Phe Ile Gln Asn Arg Asp Glu Gly Ser Glu Pro Val
   545                               550                               555                               560

Pro Ala Arg Val Pro Glu Glu Gly Leu Ala Pro Ala Pro Gly His Tyr
   565                               570                               575

Val Lys Val Arg Leu Asn Pro Val Phe Ala Cys Pro Thr Glu Glu Asp
   580                               585                               590

Ala Ala

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<210> SEQ ID NO 6
<211> LENGTH: 594
<212> TYPE: PRT
<213> ORGANISM: Aeromonas caviae

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<400> SEQUENCE: 6

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Met Ser Gln Pro Ser Tyr Gly Pro Leu Phe Glu Ala Leu Ala His Tyr
 1           5           10           15

Asn Asp Lys Leu Leu Ala Met Ala Lys Ala Gln Thr Glu Arg Thr Ala
 20           25           30

Gln Ala Leu Leu Gln Thr Asn Leu Asp Asp Leu Gly Gln Val Leu Glu
 35           40           45

Gln Gly Ser Gln Gln Pro Trp Gln Leu Ile Gln Ala Gln Met Asn Trp
 50           55           60

Trp Gln Asp Gln Leu Lys Leu Met Gln His Thr Leu Leu Lys Ser Ala
 65           70           75           80

Gly Gln Pro Ser Glu Pro Val Ile Thr Pro Glu Arg Ser Asp Arg Arg
 85           90           95

Phe Lys Ala Glu Ala Trp Ser Glu Gln Pro Ile Tyr Asp Tyr Leu Lys
100           105           110

Gln Ser Tyr Leu Leu Thr Ala Arg His Leu Leu Ala Ser Val Asp Ala
115           120           125

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Leu 130	Glu	Gly	Val	Pro	Gln	Lys 135	Ser	Arg	Glu	Arg	Leu 140	Arg	Phe	Phe	Thr
Arg 145	Gln	Tyr	Val	Ser	Ala 150	Met	Ala	Pro	Ser	Asn 155	Phe	Leu	Ala	Thr	Asn 160
Pro	Glu	Leu	Leu	Lys 165	Leu	Thr	Leu	Glu	Ser	Gly 170	Gly	Gln	Asn	Leu	Val 175
Arg	Gly	Leu	Ala 180	Leu	Leu	Ala	Glu	Asp 185	Leu	Glu	Arg	Ser	Ala	Asp	Gln
Leu	Asn 195	Ile	Arg	Leu	Thr	Asp	Glu 200	Ser	Ala	Phe	Glu	Leu 205	Gly	Arg	Asp
Leu	Ala 210	Leu	Thr	Pro	Gly	Arg 215	Val	Val	Gln	Arg	Thr 220	Glu	Leu	Tyr	Glu
Leu 225	Ile	Gln	Tyr	Ser	Pro 230	Thr	Thr	Glu	Thr	Val 235	Gly	Lys	Thr	Pro	Val 240
Leu	Ile	Val	Pro	Pro 245	Phe	Ile	Asn	Lys	Tyr 250	Tyr	Ile	Met	Asp	Met	Arg 255
Pro	Gln	Asn 260	Ser	Leu	Val	Ala	Trp	Leu 265	Val	Ala	Gln	Gly	Gln	Thr	Val
Phe	Met 275	Ile	Ser	Trp	Arg	Asn 280	Pro	Gly	Val	Ala	Gln	Ala	Gln	Ile	Asp
Leu	Asp 290	Asp	Tyr	Val	Val	Asp 295	Gly	Val	Ile	Ala	Ala 300	Leu	Asp	Gly	Val
Glu 305	Ala	Ala	Thr	Gly	Glu 310	Arg	Glu	Val	His	Gly 315	Ile	Gly	Tyr	Cys	Ile 320
Gly	Gly	Thr	Ala 325	Leu	Ser	Leu	Ala	Met	Gly 330	Trp	Leu	Ala	Ala	Arg	Arg 335
Gln	Lys	Gln 340	Arg	Val	Arg	Thr	Ala	Thr 345	Leu	Phe	Thr	Thr	Leu	Leu	Asp
Phe	Ser 355	Gln	Pro	Gly	Glu	Leu	Gly 360	Ile	Phe	Ile	His	Glu 365	Pro	Ile	Ile
Ala	Ala 370	Leu	Glu	Ala	Gln	Asn 375	Glu	Ala	Lys	Gly	Ile 380	Met	Asp	Gly	Arg
Gln 385	Leu	Ala	Val	Ser	Phe 390	Ser	Leu	Leu	Arg	Glu 395	Asn	Ser	Leu	Tyr	Trp 400
Asn	Tyr	Tyr	Ile 405	Asp	Ser	Tyr	Leu	Lys	Gly 410	Gln	Ser	Pro	Val	Ala	Phe 415
Asp	Leu	Leu	His 420	Trp	Asn	Ser	Asp	Ser 425	Thr	Asn	Val	Ala	Gly	Lys	Thr 430
His	Asn 435	Ser	Leu	Leu	Arg	Arg	Leu 440	Tyr	Leu	Glu	Asn 445	Gln	Leu	Val	Lys
Gly 450	Glu	Leu	Lys	Ile	Arg	Asn 455	Thr	Arg	Ile	Asp	Leu 460	Gly	Lys	Val	Lys
Thr 465	Pro	Val	Leu	Leu	Val 470	Ser	Ala	Val	Asp	Asp 475	His	Ile	Ala	Leu	Trp 480
Gln	Gly	Thr	Trp 485	Gln	Gly	Met	Lys	Leu	Phe 490	Gly	Gly	Glu	Gln	Arg	Phe 495
Leu	Leu	Ala 500	Glu	Ser	Gly	His	Ile 505	Ala	Gly	Ile	Ile	Asn	Pro	Pro	Ala 510
Ala	Asn 515	Lys	Tyr	Gly	Phe	Trp	His 520	Asn	Gly	Ala	Glu 525	Ala	Glu	Ser	Pro
Glu 530	Ser	Trp	Leu	Ala	Gly	Ala 535	Thr	His	Gln	Gly	Gly 540	Ser	Trp	Trp	Pro
Glu	Met	Met	Gly	Phe	Ile	Gln	Asn	Arg	Asp	Glu	Gly	Ser	Glu	Pro	Val

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545	550	555	560
Pro Ala Arg Val	Pro Glu Glu Gly	Leu Ala Pro Ala	Pro Gly His Tyr
	565	570	575
Val Lys Val Arg	Leu Asn Pro Val	Phe Ala Cys Pro	Thr Glu Glu Asp
	580	585	590
Ala Ala			
<210> SEQ ID NO 7			
<211> LENGTH: 567			
<212> TYPE: PRT			
<213> ORGANISM: Chromobacterium sp.			
<400> SEQUENCE: 7			
Met Gln Gln Phe	Val Asn Ser Leu	Ser Leu Gly Gln	Asp Gln Ser Asp
1	5	10	15
Ala Pro His Pro	Leu Thr Gly Ala	Trp Ser Gln Leu	Met Ser Gln Thr
	20	25	30
Asn Gln Leu Leu	Gln Leu Gln Ser	Ser Leu Tyr Gln	Gln Gln Leu Gly
	35	40	45
Leu Trp Thr Gln	Phe Leu Gly Gln	Thr Ala Gly Asn	Asp Ala Ser Ala
	50	55	60
Pro Ser Ala Lys	Pro Ser Asp Arg	Arg Phe Ala Ser	Pro Glu Trp Asp
	65	70	75
Glu His Pro Phe	Tyr Ser Phe Leu	Lys Gln Ser Tyr	Leu Gln Thr Ser
	85	90	95
Lys Trp Met Met	Glu Leu Val Asp	Lys Thr Gln Ile	Asp Glu Ser Ala
	100	105	110
Lys Asp Lys Leu	Ser Phe Ala Thr	Arg Gln Tyr Leu	Asp Ala Met Ala
	115	120	125
Pro Ser Asn Phe	Met Leu Thr Asn	Pro Asp Val Val	Lys Arg Ala Ile
	130	135	140
Glu Thr Gln Gly	Glu Ser Leu Val	Glu Gly Met Lys	Asn Met Met Glu
	145	150	155
Asp Ile Gln Lys	Gly His Ile Ser	Met Ser Asp Glu	Ser Lys Phe Gln
	165	170	175
Ile Gly Lys Asn	Leu Val Val Thr	Pro Gly Glu Val	Val Phe Arg Asn
	180	185	190
Glu Leu Ile Glu	Leu Ile Gln Tyr	Thr Pro Thr Thr	Glu Lys Val His
	195	200	205
Glu Lys Pro Leu	Leu Phe Val Pro	Pro Cys Ile Asn	Lys Tyr Tyr Leu
	210	215	220
Met Asp Leu Gln	Pro Asp Asn Ser	Met Val Arg His	Phe Val Gly Gln
	225	230	235
Gly Tyr Arg Val	Phe Leu Val Ser	Trp Arg Ser Ala	Val Pro Glu Met
	245	250	255
Lys Asn Phe Thr	Trp Glu Thr Tyr	Ile Glu Lys Gly	Val Phe Ala Ala
	260	265	270
Ala Glu Ala Val	Gln Lys Ile Thr	Lys Gln Pro Thr	Met Asn Ala Leu
	275	280	285
Gly Phe Cys Val	Gly Gly Val Ile	Leu Thr Thr Ala	Leu Cys Val Ala
	290	295	300
Gln Ala Lys Gly	Leu Lys Tyr Phe	Asp Ser Ala Thr	Phe Met Thr Ser
	305	310	315
Leu Ile Asp His	Ala Glu Pro Gly	Glu Ile Ser Phe	Phe Ile Asp Glu

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325					330					335					
Ala	Leu	Val	Ala	Ser	Arg	Glu	Ala	Lys	Met	Ala	Ala	Gly	Gly	Ile	Ile
			340					345					350		
Ser	Gly	Lys	Glu	Ile	Gly	Arg	Thr	Phe	Ala	Ser	Leu	Arg	Ala	Asn	Asp
		355					360					365			
Leu	Val	Trp	Asn	Tyr	Val	Val	Asn	Asn	Tyr	Leu	Leu	Gly	Lys	Thr	Pro
		370					375					380			
Ala	Pro	Phe	Asp	Leu	Leu	Tyr	Trp	Asn	Asn	Asp	Ala	Val	Asp	Leu	Pro
				390								395			400
Leu	Pro	Met	His	Thr	Phe	Met	Leu	Arg	Gln	Phe	Tyr	Ile	Asn	Asn	Ala
				405					410					415	
Leu	Ile	Thr	Pro	Gly	Ala	Ile	Thr	Leu	Cys	Gly	Val	Pro	Ile	Asp	Ile
			420					425					430		
Ser	Lys	Ile	Asp	Ile	Pro	Val	Tyr	Met	Phe	Ala	Ala	Arg	Glu	Asp	His
		435					440					445			
Ile	Val	Leu	Trp	Ser	Ser	Ala	Tyr	Ser	Gly	Leu	Lys	Tyr	Leu	Ser	Gly
		450					455					460			
Thr	Pro	Ser	Arg	Arg	Phe	Val	Leu	Gly	Ala	Ser	Gly	His	Ile	Ala	Gly
				470					475						480
Ser	Ile	Asn	Pro	Val	Thr	Lys	Asp	Lys	Arg	Asn	Tyr	Trp	Thr	Asn	Glu
				485					490					495	
Gln	Leu	Pro	Val	Asn	Pro	Glu	Glu	Trp	Leu	Glu	Gly	Ala	Gln	Ser	His
			500					505					510		
Pro	Gly	Ser	Trp	Trp	Lys	Asp	Trp	Asp	Ala	Trp	Leu	Ala	Pro	Gln	Ser
		515					520					525			
Gly	Lys	Gln	Val	Pro	Ala	Pro	Lys	Met	Leu	Gly	Ser	Lys	Glu	Phe	Pro
		530					535					540			
Pro	Leu	Gln	Pro	Ala	Pro	Gly	Ser	Tyr	Val	Leu	Ala	Lys	Ala	Met	Pro
				550					555					560	
Pro	Val	Ala	Ala	Ala	Leu	Asn									
				565											

<210> SEQ ID NO 8

<211> LENGTH: 559

<212> TYPE: PRT

<213> ORGANISM: Pseudomonas sp.

<400> SEQUENCE: 8

Met	Ser	Asn	Lys	Asn	Ser	Asp	Asp	Leu	Asn	Arg	Gln	Ala	Ser	Glu	Asn
1				5						10				15	
Thr	Leu	Gly	Leu	Asn	Pro	Val	Ile	Gly	Leu	Arg	Gly	Lys	Asp	Leu	Leu
			20						25				30		
Thr	Ser	Ala	Arg	Met	Val	Leu	Thr	Gln	Ala	Ile	Lys	Gln	Pro	Ile	His
		35					40					45			
Ser	Val	Lys	His	Val	Ala	His	Phe	Gly	Ile	Glu	Leu	Lys	Asn	Val	Met
		50				55					60				
Phe	Gly	Lys	Ser	Lys	Leu	Gln	Pro	Glu	Ser	Asp	Asp	Arg	Arg	Phe	Asn
		65			70					75				80	
Asp	Pro	Ala	Trp	Ser	Gln	Asn	Pro	Leu	Tyr	Lys	Arg	Tyr	Leu	Gln	Thr
			85						90					95	
Tyr	Leu	Ala	Trp	Arg	Lys	Glu	Leu	His	Asp	Trp	Ile	Gly	Asn	Ser	Lys
		100						105					110		
Leu	Ser	Glu	Gln	Asp	Ile	Asn	Arg	Ala	His	Phe	Val	Ile	Thr	Leu	Met
		115					120						125		

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Thr	Glu	Ala	Met	Ala	Pro	Thr	Asn	Ser	Ala	Ala	Asn	Pro	Ala	Ala	Val
130						135					140				
Lys	Arg	Phe	Phe	Glu	Thr	Gly	Gly	Lys	Ser	Leu	Leu	Asp	Gly	Leu	Thr
145					150					155					160
His	Leu	Ala	Lys	Asp	Leu	Val	Asn	Asn	Gly	Gly	Met	Pro	Ser	Gln	Val
			165						170					175	
Asp	Met	Gly	Ala	Phe	Glu	Val	Gly	Lys	Ser	Leu	Gly	Thr	Thr	Glu	Gly
			180					185					190		
Ala	Val	Val	Phe	Arg	Asn	Asp	Val	Leu	Glu	Leu	Ile	Gln	Tyr	Arg	Pro
		195					200					205			
Thr	Thr	Glu	Gln	Val	His	Glu	Arg	Pro	Leu	Leu	Val	Val	Pro	Pro	Gln
	210					215					220				
Ile	Asn	Lys	Phe	Tyr	Val	Phe	Asp	Leu	Ser	Pro	Asp	Lys	Ser	Leu	Ala
225					230					235					240
Arg	Phe	Cys	Leu	Ser	Asn	Asn	Gln	Gln	Thr	Phe	Ile	Val	Ser	Trp	Arg
			245						250					255	
Asn	Pro	Thr	Lys	Ala	Gln	Arg	Glu	Trp	Gly	Leu	Ser	Thr	Tyr	Ile	Asp
			260					265					270		
Ala	Leu	Lys	Glu	Ala	Val	Asp	Val	Val	Ser	Ala	Ile	Thr	Gly	Ser	Lys
		275					280					285			
Asp	Ile	Asn	Met	Leu	Gly	Ala	Cys	Ser	Gly	Gly	Ile	Thr	Cys	Thr	Ala
	290					295					300				
Leu	Leu	Gly	His	Tyr	Ala	Ala	Leu	Gly	Glu	Lys	Lys	Val	Asn	Ala	Leu
305					310					315					320
Thr	Leu	Leu	Val	Ser	Val	Leu	Asp	Thr	Thr	Leu	Asp	Ser	Gln	Val	Ala
			325						330					335	
Leu	Phe	Val	Asp	Glu	Lys	Thr	Leu	Glu	Ala	Ala	Lys	Arg	His	Ser	Tyr
		340						345					350		
Gln	Ala	Gly	Val	Leu	Glu	Gly	Arg	Asp	Met	Ala	Lys	Val	Phe	Ala	Trp
		355					360					365			
Met	Arg	Pro	Asn	Asp	Leu	Ile	Trp	Asn	Tyr	Trp	Val	Asn	Asn	Tyr	Leu
	370					375					380				
Leu	Gly	Asn	Glu	Pro	Pro	Val	Phe	Asp	Ile	Leu	Phe	Trp	Asn	Asn	Asp
385					390					395					400
Thr	Thr	Arg	Leu	Pro	Ala	Ala	Phe	His	Gly	Asp	Leu	Ile	Glu	Met	Phe
			405						410					415	
Lys	Asn	Asn	Pro	Leu	Val	Arg	Ala	Asn	Ala	Leu	Glu	Val	Ser	Gly	Thr
			420					425					430		
Pro	Ile	Asp	Leu	Lys	Gln	Val	Thr	Ala	Asp	Ile	Tyr	Ser	Leu	Ala	Gly
		435					440					445			
Thr	Asn	Asp	His	Ile	Thr	Pro	Trp	Lys	Ser	Cys	Tyr	Lys	Ser	Ala	Gln
	450					455					460				
Leu	Phe	Gly	Gly	Lys	Val	Glu	Phe	Val	Leu	Ser	Ser	Ser	Gly	His	Ile
465					470					475					480
Gln	Ser	Ile	Leu	Asn	Pro	Pro	Gly	Asn	Pro	Lys	Ser	Arg	Tyr	Met	Thr
			485					490						495	
Ser	Thr	Asp	Met	Pro	Ala	Thr	Ala	Asn	Glu	Trp	Gln	Glu	Asn	Ser	Thr
		500						505					510		
Lys	His	Thr	Asp	Ser	Trp	Trp	Leu	His	Trp	Gln	Ala	Trp	Gln	Ala	Glu
		515					520					525			
Arg	Ser	Gly	Lys	Leu	Lys	Lys	Ser	Pro	Thr	Ser	Leu	Gly	Asn	Lys	Ala
	530					535					540				
Tyr	Pro	Ser	Gly	Glu	Ala	Ala	Pro	Gly	Thr	Tyr	Val	His	Glu	Arg	

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545	550	555	
<210> SEQ ID NO 9 <211> LENGTH: 29 <212> TYPE: DNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Artificial DNA			
<400> SEQUENCE: 9			
ttgacaatta atcatccggc tcgtataat			29
<210> SEQ ID NO 10 <211> LENGTH: 30 <212> TYPE: DNA <213> ORGANISM: Escherichia coli			
<400> SEQUENCE: 10			
tttacacttt atgcttcgg ctcgtataat			30
<210> SEQ ID NO 11 <211> LENGTH: 30 <212> TYPE: DNA <213> ORGANISM: Escherichia coli			
<400> SEQUENCE: 11			
ttgacaatta atcatcgaac tagttaacta			30
<210> SEQ ID NO 12 <211> LENGTH: 30 <212> TYPE: DNA <213> ORGANISM: Escherichia coli			
<400> SEQUENCE: 12			
tttacacttt atgcttcgg ctcgtatgtt			30
<210> SEQ ID NO 13 <211> LENGTH: 30 <212> TYPE: DNA <213> ORGANISM: Cupriavidus necator			
<400> SEQUENCE: 13			
ttgacagcgc gtgcgttgca aggcaacaat			30
<210> SEQ ID NO 14 <211> LENGTH: 231 <212> TYPE: DNA <213> ORGANISM: Cupriavidus necator			
<400> SEQUENCE: 14			
tacgccgccc cctgaccagg aacgccgggc cagtcgccgc gtttttttat tctatagcgc			60
aattaaccgc cgtcatattg cgtcaccatg attgccggat ggccgcggcg atcccttget			120
ggaggccggt tccaagaaga tttaaagatg tcacggaatt gtcatacagg gagcatagag			180
ttcgtcttgt caaaaatttg tcattcccaa ccaatgttct ctggaggaca t			231
<210> SEQ ID NO 15 <211> LENGTH: 17 <212> TYPE: DNA <213> ORGANISM: Cupriavidus necator			
<400> SEQUENCE: 15			
agagagacaa tcaaatc			17

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<210> SEQ ID NO 16
<211> LENGTH: 828
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Artificial DNA

<400> SEQUENCE: 16
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atcctgcgca tcaagctgat cggcgtggtg ccggagtctg aagccgtgct gcacgcctcg 240
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gtggtggacc gcttctcctg caaggacaag ccgatgcgtt tcaccgacta ccagaagccg 360
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gagttttgtg tccttgtagc ttacgttttg ttaaattttc gcggcgtcta caacgaaacg 720
tagacatcgt ccccgccgtg atccggcgga gagaacagta caagacagaa aaggagcacc 780
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<210> SEQ ID NO 17
<211> LENGTH: 1100
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Artificial DNA

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gccatcggcg tgggtggccc ccccgccag cgcgagtggg ccgagcgctt cggcctgcgg 540
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gacgtggta tctcggacgt ggtcagctac ggccgcgagg tgatcgccga aggaacatc 840
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cgcattttca gcacgtgcat ggagcctgaa ctgattttcca tcgccggcat ctaccggacc 960
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gaaaaactga tcctggaagc gctgcggctc aagtaacctg ccggcctggt tcaaccagtc 1080
ggcagccgac tagtggatcc 1100

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<210> SEQ ID NO 18
<211> LENGTH: 1103
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Artificial DNA

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<400> SEQUENCE: 18

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cgatcccttg ctggaggccg gttccaagaa gatttaaaga tgtaacggaa ttgtcataca 180
gggagcatag agttcgtctt gtcaaaaatt tgcattccc aaccaatgtt ctctggagga 240
catatggcaa aaatcatcgt tgtgacctcc ggcaagggag gcgtcgcaa gaccaccacc 300
agcgccagct ttgcgcggc cctggccctg cgcggccaca agactgccgt gatcgacttc 360
gacgtcggcc tgcgcaacct tgacctgac atgggttgcg agcgccgct ggtgtaogac 420
ctgatcaacg tgggtcaggg cgaagccaac ctgcgccagg cgctgatcaa ggacaagaag 480
tgcgagaacc tgttcatect gccggcctcg cagacgcgcg acaaggacgc gtcacgcgcg 540
gaaggcgtcg agaaggtcat caacggcctg atcgagatgg atttogaatt catcatctgc 600
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gtcggcagcc gactagtgga tcc 1103

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<210> SEQ ID NO 19
<211> LENGTH: 1983
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Artificial DNA

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<400> SEQUENCE: 19

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gggagcatag agttcgtctt gtcaaaaatt tgcattccc aaccaatgtt ctctggagga 240
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cttctcgccc tccagaccgc cgacatggct gcgctgcggg atgacctcct cggccgcttt 360
gaagccaccc ccgaattctt ttccaatgac gtgattgcgc tggacctgcg cgcgctggaa 420

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ctgctggaca gccaggccc cgcgcgcagt ggcccgatc gcgccaccga ccgtgccgc	600
gaggccaggg ccgcagccgc ggcggaacag gccgcagccg accaggccgc gcgcaggaa	660
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accagacca tgctgatcga caagccgctt cgtcggggc agcaggtcta cgcgcagggc	780
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cgccatccac ctggaaggca gcgacgtggc cgacgcctat ggcgacgtgg tggaccgctt	1860
cctcggcaag gacaagccga tgcgtttcac cgactaccag aagccgggtc tgcctcccg	1920
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tcc	1983

<210> SEQ ID NO 20

<211> LENGTH: 2259

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Artificial DNA

<400> SEQUENCE: 20

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cgatcccttg ctggaggccg gttccaagaa gatttaaaga tgtaacggaa ttgtcataca	180
gggagcatag agttcgtctt gtcaaaaatt tgcattccc aaccaatgtt ctctggagga	240
catatgtccc agaagaaatc gccacgcttc gagctgcgca gtggcaacgt agacgcctc	300
cttctcgccc tccagaccgc cgacatggct gcgtgcggg atgacctct cgcgcgttt	360
gaagccacc cgcacttctt ttccaatgac gtgattgcg tggacctgc cgcgtggaa	420

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gatgacagcg aagtcgcgct tggcaccgtg atcgagacgc tggccacgct cagggcccgc	480
gccatcggcg tggtagcccg ccccgccag cgcgagtggg cgcgagcgtt cggcctgccg	540
ctgctggaca gccaggcccg ccgcggcagt ggcgccgacg gcgccaccga ccgtgccgcc	600
gaggccaggg ccgcagccgc ggcggaacag gccgcagccg accaggccgc gcgcgaggaa	660
tccatccgcg ccgcgcgcga ggcaccacc gacgcgcgcg tggccgctgc catccgccag	720
accagacca tgctgatcga caagccgctt cgtcggggcc agcaggtcta cgcgcagggc	780
gacgtggtea tcctggacgt ggtcagctac ggcgccgagg tgatcgccga aggcacacac	840
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cgaagccaac ctgcgccagg cgctgatcaa ggacaagaag tgcgagaacc tgttcacct	1380
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caacggcctg atcgagatgg atttcgaatt catcatctgc gactcgccg ccggcatcga	1500
gtcgggcgcg ctgatggcga tgtacttcgc cgacgaggcg ctgatcgtga ccaacccgga	1560
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cgcacagcaa ggccggcgacc cgatcaagga acacctgctg atcacccgct acaaccccaa	1680
gcgtgtgcat ggccggcga tctgtctgct gaccgacatc caggaaatcc tgcgcacaa	1740
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cgcacccac ctggaaggca gcgacgtggc cgacgcctat ggcgacgtgg tggaccgctt	1860
cctcggcaag gacaagccga tgcgtttcac cgactaccag aagccgggtc tgctctccg	1920
catcttcggc aacaagtaac ggtcaaggag ggctcacccc atgtcgatcc ttctcttct	1980
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ggtggcggtg atttcaagc acgtcaagat cggcgaccag gacctgcgcg tcagcctgga	2160
gcgcaggac aacctcgagg tgctcgaggt caagatcgag atccccgaga actgacctgc	2220
cggcctggtt caaccagtcg gcagccgact agtggatcc	2259

The invention claimed is:

1. A transformed microorganism, belonging to the genus *Cupriavidus*, and comprising a polyhydroxyalkanoate synthase gene, wherein expression of a minC gene and a minD gene is enhanced as compared to a wild strain of the microorganism,

wherein

the minC gene is a gene having a nucleotide sequence that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 1, or a polypeptide comprising an amino acid sequence that is at least 90% homologous to the amino acid sequence of SEQ ID NO: 1, the minD gene is a gene having a nucleotide sequence that encodes a polypeptide comprising the amino acid

sequence of SEQ ID NO: 2, or a polypeptide comprising an amino acid sequence that is at least 90% homologous to the amino acid sequence of SEQ ID NO: 2, and the polyhydroxyalkanoate synthase gene is a gene having a nucleotide sequence that encodes a polypeptide comprising the amino acid sequence of SEQ ID NOS: 4, 5, 6, 7, or 8, or a polypeptide having polyhydroxyalkanoate synthase activity and comprising an amino acid sequence that is at least 90% homologous to the amino acid sequence of SEQ ID NOS: 4, 5, 6, 7, or 8, and a size of microbial cells of the transformed microorganism that accumulates polyhydroxyalkanoate is larger compared to a size of microbial cells of the wild strain that accumulates the polyhydroxyalkanoate, and the

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transformed microorganism accumulates larger-size particles of the polyhydroxyalkanoate compared to the wild strain.

2. The transformed microorganism according to claim 1, wherein expression of a minE gene is enhanced as compared to a wild strain of the microorganism,

wherein the minE gene is a gene having a nucleotide sequence that encodes a polypeptide comprising the amino acid sequence of SEQ ID NO: 3, or a polypeptide comprising an amino acid sequence that is at least 90% homologous to the amino acid sequence of SEQ ID NO: 3.

3. The transformed microorganism according to claim 2, which is transformed *Cupriavidus necator*.

4. A method of producing a polyhydroxyalkanoate, comprising culturing the transformed microorganism of claim 2 in the presence of a carbon source.

5. The method according to claim 4, wherein the carbon source comprises an oil or a fatty acid.

6. The method according to claim 4, wherein the carbon source comprises a sugar.

7. The method according to claim 4, wherein the carbon source contains comprises carbon dioxide.

8. The method according to claim 4, wherein the polyhydroxyalkanoate is a copolymer of at least two hydroxyalkanoates.

9. The method according to claim 8, wherein the polyhydroxyalkanoate is a copolymer comprising 3-hydroxyhexanoate as a monomer unit.

10. The method according to claim 9, wherein the polyhydroxyalkanoate is a copolymer of 3-hydroxybutyrate and 3-hydroxyhexanoate.

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11. The transformed microorganism according to claim 1, which is transformed *Cupriavidus necator*.

12. A method of producing a polyhydroxyalkanoate, comprising culturing the transformed microorganism of claim 1 in the presence of a carbon source.

13. The method according to claim 12, wherein the carbon source comprises an oil or a fatty acid.

14. The method according to claim 12, wherein the carbon source comprises a sugar.

15. The method according to claim 12, wherein the carbon source comprises carbon dioxide.

16. The method according to claim 12, wherein the polyhydroxyalkanoate is a copolymer of at least two hydroxyalkanoates.

17. The method according to claim 16, wherein the polyhydroxyalkanoate is a copolymer comprising 3-hydroxyhexanoate as a monomer unit.

18. The method according to claim 17, wherein the polyhydroxyalkanoate is a copolymer of 3-hydroxybutyrate and 3-hydroxyhexanoate.

19. The transformed microorganism according to claim 1, wherein the size of microbial cells of the transformed microorganism that accumulates the polyhydroxyalkanoate is at least 2.24 μm , and the size of polyhydroxyalkanoate particles that are accumulated by the transformed microorganism is at least 1.94 μm .

20. The transformed microorganism according to claim 2, wherein the size of microbial cells of the transformed microorganism that accumulates the polyhydroxyalkanoate is at least 2.34 μm , and the size of polyhydroxyalkanoate particles that are accumulated by the transformed microorganism is at least 1.87 μm .

* * * * *